

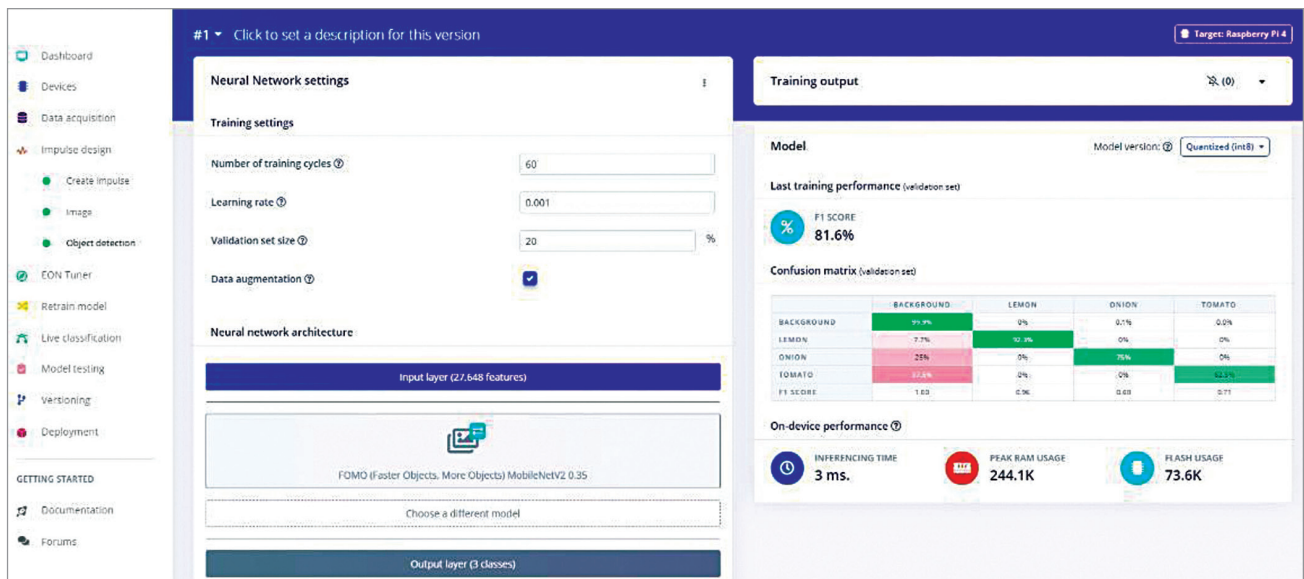


Fig. 3: Training ML model

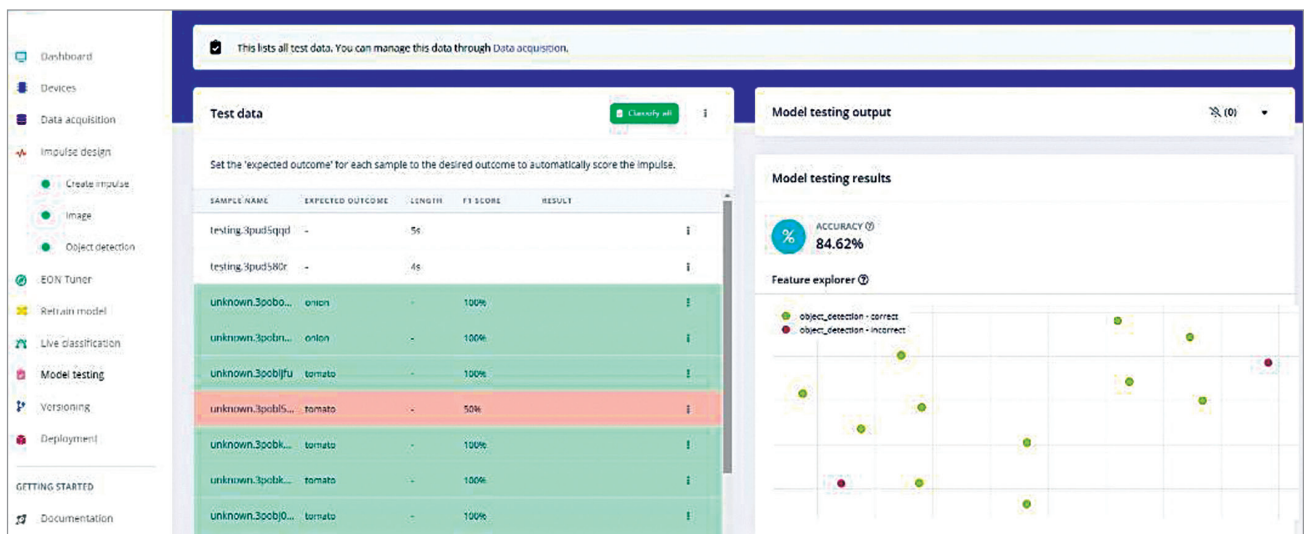


Fig. 4: Testing model

will build the project for you.

To collect pictures of the items, connect a Raspberry Pi computer and start the edgeimpulse project using Raspberry Pi and a phone camera or a laptop camera. The following steps would make the process clear:

1. Go to edgeimpulse.com, login, and start a new project. You will notice on the right top 'collect data' and below that 'Connect a device to start building your dataset.'

2. Collect a good amount of data of the objects from several angles in

several combinations. Fig. 2 shows one such dataset being collected.

3. On the left top there are three buttons (Dataset, Data source, Labeling queue). Click on Data source and point it to the camera in your smartphone (or laptop camera). Your smartphone is now ready for clicking the required pictures.

4. To create the project, go to Dashboard→Devices→Data acquisition (Impulse design→create impulse→image→object detection). After collecting a good number of

pictures (say, 200) of all the items to be segregated, divide them in 80:20 ratio for training and testing. All the images have to be labeled with a surrounding box. To avoid repetitive box for labelling, go to 'labelling queue' under 'label suggestions' and select 'Classify using YOLOv5'

The beginners may just go with the default selections in 'create impulse' and 'image' sections, to keep it simple.

5. In 'Object detection' there are several models for classifica-